

Billy Kelley

Senior Full Stack AI Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack AI Engineer with 10+ years of building enterprise applications, distributed systems, and production AI solutions. Hands-on experience delivering full-stack applications using React, Next.js, PostgreSQL, and Python integrating LLMs (Azure OpenAI, Claude) and LangChain into production workflows. Built a production RAG platform using Azure OpenAI and LangChain supporting 8,000+ concurrent users with Kafka ingestion pipelines processing 18M+ document events/month; focused on SDLC, CI/CD, observability, and performance tuning to sustain 99.95% uptime. Experienced delivering AI-enabled solutions for healthcare clients and translating business requirements into scalable, secure systems.

EXPERIENCE

Senior Software Engineer, *Blackbaud* 11/2025 – 06/2026 | Charleston, SC

- Enhanced an AI transcript summarization workflow using Claude (LLM integration), standardizing summaries across thousands of customer records to improve downstream search and supportability.
- Led implementation of a centralized Single Sign-On platform with React, Next.js, Express, OAuth 2.0 and FusionAuth, replacing fragmented authentication across multiple applications and improving security and developer velocity.
- Modernized GitHub Actions and Jenkins CI/CD pipelines supporting multiple enterprise authentication services and 8+ microservices, reducing build/deploy time and increasing release reliability.
- Leveraged Advanced Data Analytics to build and optimize machine learning models for AI applications.
- Built Prometheus and Grafana dashboards to provide engineers immediate visibility into production health on Linux systems hosted on AWS, supporting 99.95% uptime and enabling faster root-cause analysis.

AI Engineer, *Pryon* 02/2024 – 10/2025 | Raleigh, NC

- Developed production RAG applications using Azure OpenAI and LangChain, combining vector search with Azure AI Search and BM25 retrieval through Elasticsearch to support 8,000+ concurrent users; implemented LLM prompting and API integrations for robust inference.
- Designed scalable Kafka ingestion pipelines processing over 8 million document events per month, automating document parsing, LLM inference, embedding generation, and search indexing for production knowledge systems.
- Integrated Pinecone for vector storage alongside Snowflake and Databricks for enterprise data processing, enabling scalable retrieval, LLM enrichment, and analytics across production AI applications.
- Instrumented observability and supported performance tuning and debugging for production RAG services to maintain system stability and meet operational SLAs.
- Performed comprehensive QA testing on AI/ML algorithms and data pipelines to guarantee optimal performance

Senior Full Stack Web Engineer, *Carolina Web Design Services, LLC* 03/2021 – 12/2023 | Sanford, NC

- Delivered 15+ production web applications using React, Next.js, Node.js, TypeScript, and REST APIs, translating business requirements into scalable, maintainable solutions for healthcare, retail, professional services, and e-commerce clients.
- Refactored and standardized shared Angular components across 8 production real estate web applications, improving code maintainability, consistency, and long-term scalability while reducing duplicate development effort.
- Developed a CRM-integrated newsletter backend server using .NET, Go and GraphQL supporting 12,000+ subscribers with automated audience synchronization, campaign management, and secure JWT authentication.

Web Developer, *Hash Triangle* 06/2018 – 02/2021 | Raleigh, NC

- Maintained and enhanced more than 15 production web applications using React, Next.js, HTML5, Tailwind CSS and PostgreSQL, delivering new features and resolving production issues throughout the Software Development Lifecycle while supporting front-end development and application scalability.

- Collaborated with UI/UX designers, product owners and engineering teams in an Agile environment to gather requirements, translate business requirements into technical solutions, produce clear technical documentation, optimize SQL Server and PostgreSQL queries, and improve application performance using Redis caching and Google Cloud Platform services.
- Expanded automated testing with Jest and Cypress while supporting Jenkins deployment pipelines, contributing to more reliable releases across 20+ production applications through continuous integration, quality assurance and deployment best practices.

EDUCATION

Bachelor of Computer Science, Fayetteville State University

04/2006 – 09/2010 | Fayetteville, NC

SKILLS

AI & Machine Learning RAG, LangChain, LLM, Azure OpenAI, Claude Code, OpenAI Codex, Vector Search, Elasticsearch, Vector Databases, Large Language Models	Backend & APIs C#, .NET, Node.js, Go, Python, REST API, GraphQL, Microservices, Kafka	Compliance & Security JWT, OAuth 2.0, HIPAA, SoC, FedRAMP	Frontend React, Next.js, TypeScript, Angular, HTML5, CSS3, JavaScript
Cloud & DevOps Azure, AWS, Docker, Kubernetes, GitHub Actions, Azure DevOps, GCP, CI/CD, SDLC	Data & Storage MySQL, SQL Server, PostgreSQL, Redis, Data Structures, Systems Design, Snowflake, Databricks	Observability & Testing Prometheus, Grafana, Jest, Cypress, Evaluation Authoring	Tools Git, Visual Studio, Jira, Swagger/Open API, OpenTelemetry

CERTIFICATES

AWS Certified Solutions Architect Amazon Web Services	Azure AI Engineer Associate Microsoft	Certified Kubernetes Application Developer The Linux Foundation
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STRENGTH

- Problem Solving**
Quickly analyzes complex technical challenges and delivers practical, scalable solutions.
- AI/ML Systems Design**
Designs secure, high-performance software architectures that scale with business needs.
- Cross-Functional Technical Leadership**
Collaborates effectively across engineering, design, product and business teams to deliver impactful solutions.